Final note: Intermediate forgetting

Standalone theoretical note

Here we gather the results of intermediate forgetting rates. First we provide an equivalent condition based on the fixed policy of using always product 1 where the maximum advantage is positive and hence, product 2 is used. This result allows us to characterize the conditions under which product 2 is used, using a policy that is easier to implement.

Proposition 1 (Characterization of positive advantage)**.** *Let U^1_ρ be the value of always choosing product 1. For $\rho \in (0,1)$, we have $\mathcal{A}_\rho > 0$ if and only if*

$$\max_{z \in \mathcal{K}_\rho} \left[U^1_\rho(T^s_\rho z) - U^1_\rho(T^f_\rho z) \right] > \frac{\Delta c}{\gamma \Delta p}. \tag{1}$$

Second, we prove that, when the maximum advantage is positive, product 2 is used not only once but infinitely often. This result is for every $\rho \in (0,1)$ where $\mathcal{A}_\rho > 0$.

Proposition 2 (Persistent use of product 2)**.** *If $\mathcal{A}_\rho > 0$, then there is a constant $\eta > 0$, such that*

$$\liminf_{N \to \infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{i_n = 2\} \geq \eta \qquad a.s, \tag{2}$$

where i_n is the product chosen at the nth interaction with the provider A. In particular, product 2 is used infinitely often.

Now we can think about the information that the fix policy of using product 2 gives us. Consider the following notation:

$$a_\rho(z) = -\Delta c + \gamma \Delta p \left[V_\rho(T^s_\rho z) - V_\rho(T^f_\rho z) \right] \tag{3}$$

$$a^1_\rho(z) = -\Delta c + \gamma \Delta p \left[U^1_\rho(T^s_\rho z) - U^1_\rho(T^f_\rho z) \right] \tag{4}$$

$$a^2_\rho(z) = -\Delta c + \gamma \Delta p \left[U^2_\rho(T^s_\rho z) - U^2_\rho(T^f_\rho z) \right] \tag{5}$$

Then $\mathcal{A}_\rho = \max_{z \in \mathcal{K}_\rho} a_\rho(z)$ and $\mathcal{A}^1_\rho = \max_{z \in \mathcal{K}_\rho} a^1_\rho(z)$. We define

$$\mathcal{A}^2_\rho = \min_{z \in \mathcal{K}_\rho} a^2_\rho(z) \tag{6}$$

The proposition shows that neither product is optimal throughout $\mathcal{K}_\rho$: the provider's optimal product choice depends on the state. It also guarantees an interior state at which both products are optimal. However, it does not establish a threshold structure, since it does not determine the number or ordering of the regions in which each product is optimal.

Lemma 1. *For $\rho \in (\frac{1}{2}, 1)$, if $\mathcal{A}^1_\rho > 0$ and $\mathcal{A}^2_\rho < 0$, then there exists $w, y \in \mathcal{K}_\rho$ such that using product 2 at w and using product 1 at y is optimal. Moreover, there exists $z^* \in \mathcal{K}_\rho$ such that $a_\rho(z^*) = 0$.*

The same strict advantage conditions also imply that both products are used persistently. This conclusion does not require connectedness of $\mathcal{K}_\rho$.

Corollary 1 (Persistent use of both products)**.** *Maintain the assumptions of the forgetting model and fix $\rho \in (0,1)$. Suppose that $\mathcal{A}_\rho^1 > 0$ and $\mathcal{A}_\rho^2 < 0$. Then there exist constants $\eta_1, \eta_2 > 0$ such that, for every initial state in $\mathcal{Z}$ and under any optimal policy,*

$$\liminf_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{i_n = j\} \geq \eta_j \qquad \text{almost surely,} \qquad j \in \{1,2\}, \tag{7}$$

where i_n denotes the product chosen at the nth interaction with provider A. In particular, both products are used infinitely often almost surely.

Lastly, before moving to the fluid model, we present a result that identifies the cost region where both products are strictly optimal at some states.

For $i \in \{1,2\}$, let

$$H_\rho^i(z) = \mathbb{E}_z^i \left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbf{1}\{y_t = A\} \right] \tag{8}$$

be the expected discounted number of purchases from seller A when product i is always used. Here t counts calendar periods, and $\mathbb{E}_z^i$ denotes expectation starting from z under this fixed policy.

Proposition 3 (Cost region with both products strictly optimal at some states)**.** *Fix $\rho \in (1/2, 1)$, $\gamma \in (0,1)$, $p_0 \in (0,1)$, $0 < p_1 < p_2 < 1$, and $0 \leq c_1 < R$. Define*

$$\underline{c}_2 = \frac{c_1 + \gamma \Delta p \, R \min_{z \in \mathcal{K}_\rho} \left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right]}{1 + \gamma \Delta p \min_{z \in \mathcal{K}_\rho} \left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right]},$$

$$\bar{c}_2 = c_1 + \gamma \Delta p \, (R - c_1) \max_{z \in \mathcal{K}_\rho} \left[H_\rho^1(T_\rho^s z) - H_\rho^1(T_\rho^f z) \right].$$

These bounds do not depend on c_2. For any $c_2 \in (c_1, R)$,

$$\mathcal{A}_\rho^1 > 0 \text{ and } \mathcal{A}_\rho^2 < 0 \quad \Longleftrightarrow \quad \underline{c}_2 < c_2 < \bar{c}_2. \tag{9}$$

Thus $(\underline{c}_2, \bar{c}_2) \cap (c_1, R)$ is exactly the cost region where each product is strictly optimal at some state in $\mathcal{K}_\rho$. For every cost in this region, there exists $x^* \in (0,1)$ such that $a_\rho(x^*, 1-x^*) = 0$.

1 Fluid Model

Derivation of the fluid model

Assume

$$0 < p_1 < p_2 < 1, \quad 0 \leq c_1 < c_2 < R, \quad 0 < p_0 < 1, \quad 0 < \rho, \gamma < 1.$$

As in the fluid progress note, we use the normalized coordinate $x = (1-\rho)S \in [0,1]$ and write

$$\lambda = -\log \rho, \qquad \delta = -\log \gamma, \qquad \Delta p = p_2 - p_1, \qquad \Delta c = c_2 - c_1.$$

On the mature boundary, the probability of a purchase is

$$D(x) = 1 - I_{p_0}\left(1 + \frac{x}{1-\rho}, 1 + \frac{1-x}{1-\rho}\right).$$

Here $I_z(a,b)$ is the regularized incomplete beta function. Under product i, a purchase yields profit $R - c_i$ and succeeds with probability p_i. If no purchase occurs, the state stays at x.

Fluid Bellman for V

Write V for the value when the product is chosen optimally. Following the fluid progress note, consider the transitions over a small time step dt:

$$T^s_{dt}(x) = \rho^{dt}x + (1-\rho^{dt}), \qquad T^f_{dt}(x) = \rho^{dt}x.$$

Over this step, the profit per purchase under product i is $(R-c_i)dt$ and the discount factor is γ^{dt}. We keep $D(x)$ fixed at the original ρ as $dt \to 0$. Since

$$\begin{aligned}\rho^{dt} &= e^{-\lambda dt} = 1 - \lambda dt + o(dt),\\ \gamma^{dt} &= e^{-\delta dt} = 1 - \delta dt + o(dt),\end{aligned}$$

the state increments are

$$\begin{aligned}T^s_{dt}(x) - x &= \lambda(1-x)dt + o(dt),\\ T^f_{dt}(x) - x &= -\lambda x dt + o(dt).\end{aligned}$$

Under product i, the expected change in the state is

$$D(x)\big[p_i\lambda(1-x) + (1-p_i)(-\lambda x)\big]dt + o(dt) = D(x)\lambda(p_i - x)dt + o(dt).$$

Thus the fluid state equation and profit rate are

$$\dot{x} = D(x)\lambda(p_i - x), \qquad D(x)(R-c_i). \tag{10}$$

To keep the notation simple, we use V at both the discrete and fluid stages of the following formal first-order derivation. The Bellman equation over dt is

$$\begin{aligned}V(x) = (1-D(x))\gamma^{dt}V(x)\\ + D(x)\max_{i\in\{1,2\}}\left\{(R-c_i)dt + \gamma^{dt}\left[p_iV(T^s_{dt}(x)) + (1-p_i)V(T^f_{dt}(x))\right]\right\}.\end{aligned} \tag{11}$$

Taylor expansion gives

$$\begin{aligned}V(T^s_{dt}(x)) &= V(x) + \lambda(1-x)V'(x)dt + o(dt),\\ V(T^f_{dt}(x)) &= V(x) - \lambda x V'(x)dt + o(dt).\end{aligned}$$

Combining the success and failure terms,

$$p_iV(T^s_{dt}(x)) + (1-p_i)V(T^f_{dt}(x)) = V(x) + \lambda(p_i - x)V'(x)dt + o(dt).$$

Substituting this and $\gamma^{dt} = 1 - \delta dt + o(dt)$ into (11) and collecting terms,

$$V(x) = V(x) + \Big[D(x)\max_{i\in\{1,2\}}\{R - c_i + \lambda(p_i - x)V'(x)\} - \delta V(x)\Big]dt + o(dt).$$

Cancelling $V(x)$, dividing by dt, and letting $dt \to 0$ yields

$$\delta V(x) = D(x)\max_{i\in\{1,2\}}\{R - c_i + \lambda(p_i - x)V'(x)\}. \tag{12}$$

To obtain the discounted profit representation, let $i(t)$ denote any admissible product choice and let $x(t)$ follow (10), starting at $x(0) = x$. Since (12) takes the maximum over products, wherever V is differentiable we have

$$\delta V(x(t)) \geq D(x(t))(R - c_{i(t)}) + V'(x(t))\dot{x}(t).$$

Multiplying by $e^{-\delta t}$ and using the chain rule gives

$$\begin{aligned}\frac{d}{dt}\big[e^{-\delta t}V(x(t))\big] &= e^{-\delta t}\big[V'(x(t))\dot{x}(t) - \delta V(x(t))\big] \\ &\leq -e^{-\delta t}D(x(t))(R - c_{i(t)}).\end{aligned}$$

Integrating from 0 to T,

$$e^{-\delta T}V(x(T)) - V(x) \leq -\int_0^T e^{-\delta t}D(x(t))(R - c_{i(t)})\,dt.$$

Rearranging,

$$V(x) \geq \int_0^T e^{-\delta t}D(x(t))(R - c_{i(t)})\,dt + e^{-\delta T}V(x(T)).$$

Because V is bounded and $\delta > 0$, the terminal term tends to zero as $T \to \infty$. Thus $V(x)$ is at least the discounted profit of every admissible policy. For an admissible policy attaining the Bellman maximum along its trajectory, these inequalities become equalities. This includes holding mixtures of tied maximizing products, as described below. The chain-rule calculation applies where V is differentiable; at corners, dynamic programming supplies the corresponding argument. Consequently, the fluid value is the largest discounted profit over these trajectories:

$$V(x) = \sup \int_0^\infty e^{-\delta t}D(x(t))(R - c_{i(t)})\,dt, \qquad x(0) = x. \tag{13}$$

We also allow mixtures: both the drift and the profit rate are averaged using the same product fractions. In particular, at $x \in [p_1, p_2]$, the product-2 fraction

$$\frac{x - p_1}{p_2 - p_1}$$

makes the average drift zero and holds the state at x. Maximizing over mixtures gives the same Bellman equation, because both drift and profit depend linearly on the fractions.

Where V is differentiable, product 2 is strictly preferred when

$$\begin{aligned}R - c_2 + \lambda(p_2 - x)V'(x) &> R - c_1 + \lambda(p_1 - x)V'(x) \\ \iff \qquad V'(x) &> \frac{\Delta c}{\lambda \Delta p}.\end{aligned}$$

Transformation

We want to rewrite the objective so that the product choice enters only through the movement of the state. To find the right adjustment, compare the two products at the same state x. Choosing product 2 reduces the current profit rate by $D(x)\Delta c$ and increases the state drift by $\lambda D(x)\Delta p$. Their ratio is

$$\frac{D(x)\Delta c}{\lambda D(x)\Delta p} = \frac{\Delta c}{\lambda \Delta p}.$$

This ratio is independent of x. It suggests crediting each extra unit of state accumulation at the rate $\Delta c/(\lambda \Delta p)$, so that the extra drift exactly offsets the extra product cost.

To see the cancellation explicitly, the two products satisfy

$$c_i = c_1 + \frac{\Delta c}{\Delta p}(p_i - p_1), \qquad i \in \{1, 2\}.$$

Adding the state credit to the profit rate and using $\dot{x} = \lambda D(x)(p_i - x)$ gives

$$\begin{aligned} D(x)(R - c_i) + \frac{\Delta c}{\lambda \Delta p}\dot{x} &= D(x)\left[R - c_i + \frac{\Delta c}{\Delta p}(p_i - x)\right] \\ &= D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right]. \end{aligned}$$

The last expression is the same for either product. Rearranging,

$$D(x)(R - c_i) = D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right] - \frac{\Delta c}{\lambda \Delta p}\dot{x}.$$

The first term now depends only on the state; the product choice affects the second term through $\dot{x}$. The same identity holds for mixtures, since both the drift and the profit are averaged with the same fractions.

Next, remove the term involving $\dot{x}$ from the discounted objective. For a trajectory starting at $x(0) = x$, integration by parts gives

$$\begin{aligned} \int_0^T e^{-\delta t}\dot{x}(t)\,dt &= \left[e^{-\delta t}x(t)\right]_0^T + \delta \int_0^T e^{-\delta t}x(t)\,dt \\ &= e^{-\delta T}x(T) - x + \delta \int_0^T e^{-\delta t}x(t)\,dt. \end{aligned}$$

Since $x(T) \in [0, 1]$ and $\delta > 0$, the terminal term vanishes as $T \to \infty$. Therefore,

$$-\frac{\Delta c}{\lambda \Delta p}\int_0^\infty e^{-\delta t}\dot{x}(t)\,dt = \frac{\Delta c}{\lambda \Delta p}x - \frac{\delta \Delta c}{\lambda \Delta p}\int_0^\infty e^{-\delta t}x(t)\,dt.$$

Combining this with the rewritten profit rate identifies the payoff that remains inside the integral:

$$F(x) = D(x)\left[R - c_1 + \frac{\Delta c}{\Delta p}(p_1 - x)\right] - \frac{\delta \Delta c}{\lambda \Delta p}x. \tag{14}$$

Thus the first term in F is the common payoff obtained after crediting state accumulation. The second is the discount correction generated by integration by parts. This is how the definition of F is obtained.

For every admissible policy, its discounted profit equals the initial state credit plus the discounted integral of F along its trajectory. Taking the supremum over policies yields

$$V(x) = \frac{\Delta c}{\lambda \Delta p}x + \sup \int_0^\infty e^{-\delta t}F(x(t))\,dt. \tag{15}$$

For a given initial state, the first term is the same for every policy, so the transformation preserves their ranking.

For the proof, write $W(x) = V(x) - \Delta c\, x/(\lambda \Delta p)$. Then W is the value of maximizing the discounted integral of F. Where the derivative exists,

$$V'(x) = W'(x) + \frac{\Delta c}{\lambda \Delta p}.$$

Substituting into (12) and using the same cancellation gives its control equation:

$$\delta W(x) = F(x) + \max_{i=1,2}\{\lambda D(x)(p_i - x)W'(x)\}. \tag{16}$$

Where the derivative exists, product 2 is strictly optimal precisely when $W' > 0$, and product 1 is strictly optimal when $W' < 0$. The transformed flow payoff is the same for both products at a given state. Products determine the direction and speed of movement through the graph of F, which is why studying the shape of this single function helps us characterize the optimal policy.

The following result proofs that the set where $F'(x) > 0$ is an interval, which will be a sufficient condition (among other requirements) to proof that the set where $W' > 0$ is an interval.

Proposition 4 (Shape of the transformed payoff)**.** *Assume $D'(x) > 0$ and D' is log-concave on $[0,1]$. Then the set $\{x \in [0,1] : F'(x) > 0\}$ is an interval, possibly empty or meeting a boundary. If nonempty, F decreases before this interval, increases within it, and decreases after it, with absent segments omitted. If it is empty, product 1 is optimal everywhere.*

Now we can prove that under ceratin conditions, the set where $W'(x) > 0$ is an interval.

Theorem 1 (A single interval of use of product 2)**.** *Suppose $0 < p_1 < p_2 < 1$, and $\delta, \lambda > 0$. Allow relaxed controls, so that any state in $[p_1, p_2]$ can be held fixed. Suppose that, for some $0 \le a < m \le 1$,*

$$F'(x) < 0 \quad \text{on } (0,a) \cup (m,1), \qquad F'(x) > 0 \quad \text{on } (a,m), \tag{17}$$

with any interior zeros of F' being simple. Then an optimal policy can be selected which uses product 2 on at most one interval and product 1 outside it. Thus, as the state increases, the only nonconstant action patterns are

$$1 \to 2, \qquad 2 \to 1, \qquad 1 \to 2 \to 1. \tag{18}$$

At an attractive threshold in (p_1, p_2), the optimal policy may use the mixture that holds the state fixed.

The monotone cases are immediate. If F is nonincreasing, product 1 is optimal everywhere. If F is nondecreasing, product 2 is optimal everywhere.

An upper bound on the number of thresholds does not guarantee that any threshold exists. The following fixed-policy tests distinguish the two constant-policy cases. Write U_ρ^i for the fluid value obtained by always using product i:

$$U_\rho^i(x) = (R - c_i) \int_0^\infty e^{-\delta t} D(x_i(t;x))\, dt.$$

Corollary 2 (Existence of a nonconstant threshold policy)**.** *If we have*

$$\max_{x \in [0,1]} \{-\Delta c + \lambda \Delta p (U_\rho^1)'(x)\} > 0,$$
$$\min_{x \in [0,1]} \{-\Delta c + \lambda \Delta p (U_\rho^2)'(x)\} < 0.$$

then there is an optimal policy with either one or two interior thresholds, with one of the three patterns in Theorem 1.

A Shape of the exact Beta demand

Lemma 2 (Shape of the exact Beta demand)**.** *Let* $0 < p_0 < 1$*,* $0 < \rho < 1$*, and* $n = (1-\rho)^{-1}$*. For*

$$D(x) = 1 - I_{p_0}\big(1 + nx, 1 + n(1-x)\big), \qquad 0 \le x \le 1,$$

we have $D'(x) > 0$ *and*

$$D'(x)D'''(x) - [D''(x)]^2 < 0.$$

In particular, D' *is strictly log-concave. The endpoint derivatives are understood through the smooth extension of* D*.*